

SAI VENKATA NAGENDRA ABHAY RAJ GURRAM

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PROFESSIONAL SUMMARY

Recent Master's graduate in Business Analytics and Artificial Intelligence with hands-on experience building Python-based machine learning, NLP, clustering, forecasting, recommendation, and decision intelligence workflows. Skilled in Python, SQL, Pandas, NumPy, Scikit-learn, HuggingFace Transformers, LangChain, OpenAI API, Power BI, Tableau, and AWS fundamentals. Experienced in model training, validation, evaluation, dashboarding, and translating ambiguous business problems into analytical and ML-driven solutions. Interested in production ML systems for forecasting, optimization, recommendation, risk assessment, model monitoring, inference optimization, and supply chain decision automation.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, Data Cleaning, Data Preprocessing, Feature Engineering, Data Transformation

Machine Learning: Scikit-learn, Supervised Learning, Unsupervised Learning, Classification, Clustering, Forecasting, Recommendation Systems, Collaborative Filtering, Predictive Modeling, Model Training, Model Validation, Model Evaluation, Cross-Validation, Hyperparameter Tuning

Deep Learning & NLP: HuggingFace Transformers, BERT, Tokenization, Embeddings, NLP Pipelines, Text Classification, Transformer Models, CNN Fundamentals, PyTorch Exposure, TensorFlow Exposure

Production ML & Optimization Interests: Model Performance Monitoring, Production Model Ownership, Alerting Concepts, Inference Optimization Interest, Quantization-Aware Training Interest, Model Compression Interest, Distillation Interest, Edge AI Interest, GPU/TPU/NPU Deployment Awareness

AI Systems & Tools: LangChain, OpenAI API, RAG-style Workflows, Pinecone, FastAPI, Streamlit, Git, Agile Development, Modular Code, Reproducible Workflows

BI & Visualization: Power BI, Tableau, Matplotlib, Plotly, Dashboards, KPI Reporting, Business Insights

Cloud & Certification: AWS Cloud Practitioner, AWS Fundamentals

PROJECT EXPERIENCE

NovaMind – Multi-Agent AI Decision Intelligence Platform

- Built a multi-agent AI platform using Python, LangChain, and OpenAI API to analyze incidents and generate structured decision-support reports.
- Designed agentic workflows for root-cause reasoning, evidence extraction, risk assessment, and narrative report generation.
- Translated ambiguous operational problems into modular AI workflows with structured outputs and human-readable recommendations.
- Developed pipeline logic for raw incident input processing, analytical reasoning, report generation, and decision-ready insight delivery.

Twitter Sentiment Analysis using BERT

- Fine-tuned a BERT Transformer model on 1.6M tweets for sentiment classification, achieving approximately 89% accuracy.

- Built an end-to-end NLP pipeline covering preprocessing, tokenization, model training, validation, evaluation, and result interpretation.
- Applied HuggingFace Transformers and Python-based experimentation to evaluate model performance and classification behavior.
- Created visual reporting outputs to communicate model results, sentiment trends, and performance insights.

Formula 1 Strategy Analysis using Clustering

- Extracted and analyzed 3 seasons of Formula 1 race data using the FastF1 library to identify pit-stop, tire, and race strategy patterns.
- Applied K-Means clustering to segment race outcomes into 3 strategy profiles based on structured performance and timing features.
- Built Python workflows for feature engineering, model evaluation, visualization, and data-driven strategy interpretation.
- Translated heterogeneous racing datasets into actionable insights around operational decision-making and performance trade-offs.

Netflix Recommendation System Optimization

- Built a collaborative filtering recommendation model using a movie ratings dataset to improve personalized content recommendation logic.
- Performed exploratory data analysis, feature engineering, model evaluation, and recommendation relevance analysis.
- Developed Python-based analytical workflows for user-item interaction analysis and recommendation performance interpretation.
- Created dashboard-style reporting outputs to summarize recommendation insights, engagement patterns, and optimization opportunities.

Rationale Agent – AI Incident Intelligence System

- Developed an AI-assisted IT operations incident reasoning system using Python and LangChain.
- Built workflows to parse unstructured incident information, identify probable root causes, and generate structured diagnostic reports.
- Applied model-assisted reasoning, evidence comparison, and output validation to support operational issue resolution.

PROFESSIONAL EXPERIENCE

Quality Assurance Intern

Jan 2022 – Jun 2022

Laurus Labs, Visakhapatnam, India

- Performed structured data validation, documentation review, and process verification within regulated pharmaceutical QA workflows.
- Reviewed process records, SOP documentation, and reporting templates to support audit readiness and data integrity.
- Collaborated with cross-functional teams to identify inconsistencies, improve reporting accuracy, and maintain documentation standards.

EDUCATION

Stevens Institute of Technology

May 2026

Master's in Business Analytics & Artificial Intelligence | GPA: 3.51

Relevant Coursework: Machine Learning, Data Mining, Web Mining, Data Management, Optimization & Process Analytics, Multivariate Data Analysis

GITAM Deemed University

Bachelor of Pharmacy

CERTIFICATIONS & ADDITIONAL INFORMATION

AWS Certified Cloud Practitioner

Apple Search Ads Certification | Bank of America Investment Banking Simulation

Publication: UV Analysis of Ritonavir | Languages: English, Telugu